Federal Aviation Administration

Initial En Route Qualification Training

Lesson 11
Route Assignments

Course 50148001

LESSON PLAN DATA SHEET

COURSE NAME:	INITIAL EN ROUTE QUALIFICATION TRAINING
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LESSON TITLE:	ROUTE ASSIGNMENTS
DURATION:	3+30 HOURS
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HANDOUT(S):	NONE
EXERCISE(S)/ ACTIVITY(S):	ACTIVITY: FIX RADIAL DISTANCE EXERCISE: ALTITUDE STRATUM
END-OF-LESSON TEST:	YES
PERFORMANCE TEST:	NONE
MATERIALS:	NONE
OTHER PERTINENT INFORMATION:	

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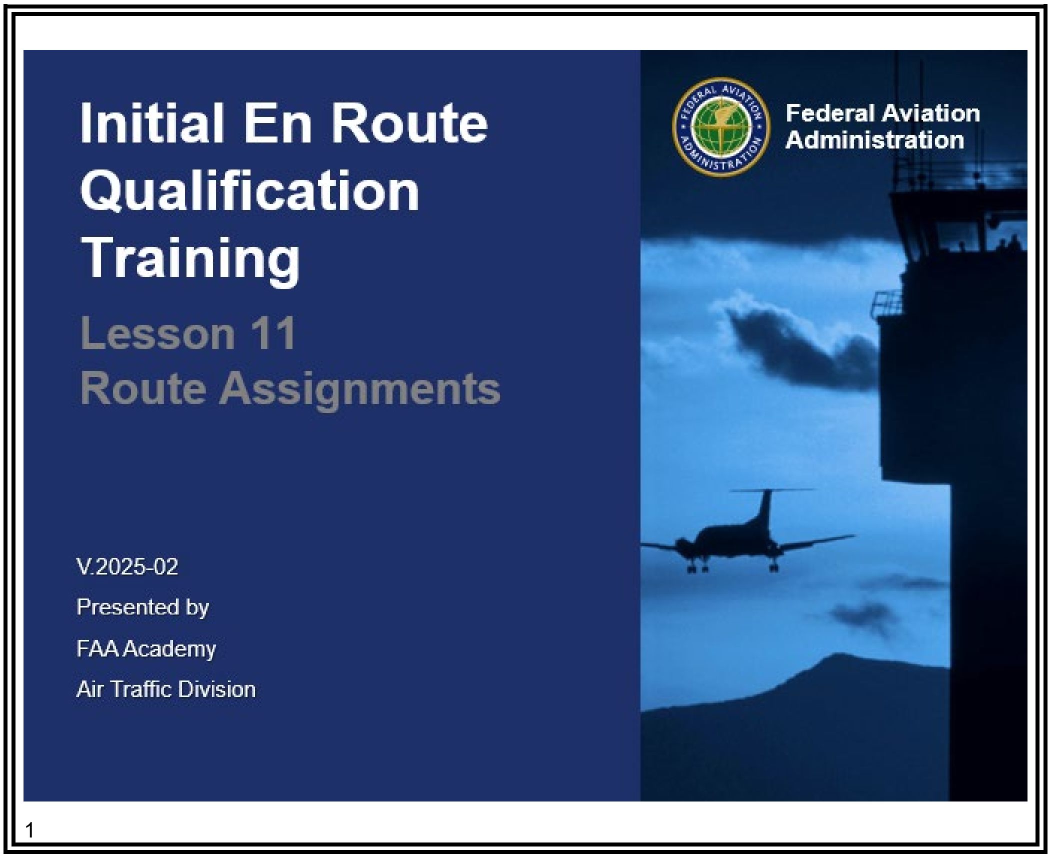
INTRODUCTION



As you proceed through this course, you continue to add skills that are required of all air traffic controllers. This lesson on IFR clearances and route assignment covers critical air traffic control functions.

Mastery of these skills will ensure that pilots have **no** doubt as to what you expect them to do. You will use skills learned in previous lessons to accurately issue clearances and route assignments.

Continued on next page

ROUTE ASSIGNMENTS

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In earlier lessons, you were introduced to Air Traffic Control (ATC) clearances and the role they play in the ATC system. ATC clearances are employed by all controllers to maintain a safe, orderly, and expeditious flow of air traffic. In this lesson, we will focus on ATC clearances as they apply in the en route environment.

Purpose

This lesson covers IFR clearance items, route/altitude amendments, and clearance prefixes.

INTRODUCTION *(Continued)*

Lesson Objectives

LESSON OBJECTIVES

- On an End-of-Lesson Test and in accordance with FAA Order JO 7110.65, you will identify:

  – Clearance and route assignment procedures for IFR aircraft
  – Phraseology for issuing selected clearances
  – Fix radial distance locations

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CLEARANCE PREFIXES

Prefix
JO 7110.65,
pars. 4-2-2, 4-2-4

✈
Phraseology

⊙ Prefix a clearance, information, or a request for information which will be relayed to an aircraft through a non-ATC facility by stating:

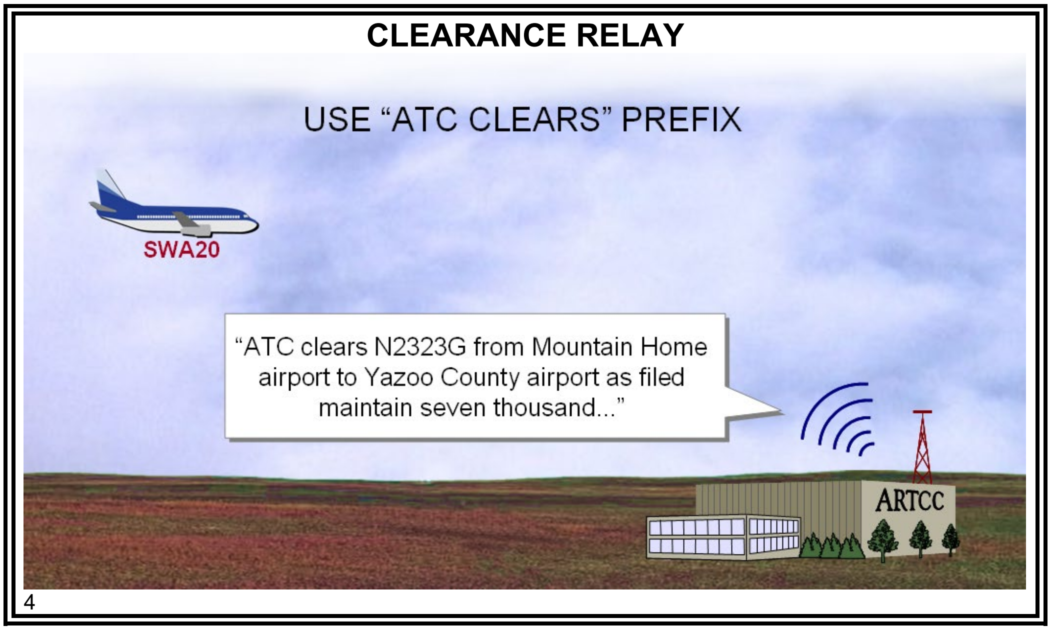
- "ATC clears" (Clearance)
- "ATC advises" (Information)

  Example: Weather information, delays at destination, etc.

- "ATC requests" (Request)

  Example: Requests for flight conditions, altitude, etc.

✈
Phraseology Example



⊙ FDU **must** prefix a clearance with the appropriate phrase:

- "ATC clears"
- "ATC advises"
- "ATC requests"

✈
Phraseology

⊙ Relay clearances verbatim.

CLEARANCE PREFIXES *(Continued)*

Knowledge Check

KNOWLEDGE CHECK

QUESTION: What prefix phrase should you use to relay a clearance to an aircraft through another aircraft?

A. "ATC clears."

B. "ATC relays."

C. "ATC advises."

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ROUTE ASSIGNMENT

Definitions
JO 7110.65, Pilot/Controller Glossary; JO 7110.10, par. 6-3-3

An **airway** is a Class E airspace area established in the form of a corridor, the centerline of which is defined by radio navigational aids; e.g., V278.

A **fix radial distance (FRD)** is a geographical position determined by a fix (up to 5 characters), an azimuth from the fix (3 digits in degrees magnetic), and a distance from the fix in nautical miles (3 digits).

VORTAC, REPORTING POINT, ETC.

Examples: SQS270040
HATER045012

FIX AZIMUTH DISTANCE

A **route** is a defined path consisting of one or more courses in a horizontal plane, which aircraft traverse over the surface of the earth.

A **jet route** is a route designed to serve aircraft operations from 18,000 feet MSL up to and including flight level 450. The routes are referred to as “J” routes with numbering to identify the designated route; e.g., J35.

A **Q Route** is an Area Navigation (RNAV) route published for use in the United States.

A **vector** is a heading issued to an aircraft to provide navigational guidance by radar.

A **Preferential Arrival Route (PAR)** is a specific arrival route from an appropriate en route point to an airport or terminal area. **It may be included in a Standard Terminal Arrival (STAR) or a Preferred IFR Route.** The abbreviation “PAR” is used primarily within the ARTCC and should **not** be confused with the abbreviation for Precision Approach Radar.

A **Standard Terminal Arrival (STAR)** is a pre-planned Instrument Flight Rule (IFR) air traffic control arrival procedure published for pilot use in graphic and/or textual form. STARs provide transition from the en route structure to an outer fix or an instrument approach fix/arrival waypoint in the terminal area.

A **Standard Instrument Departure (SID)** is a preplanned instrument flight rule (IFR) air traffic control (ATC) departure procedure printed for pilot/controller use in graphic form to provide obstacle clearance and a transition from the terminal area to the appropriate en route structure. SIDs are primarily designed for system enhancement to expedite traffic flow and to reduce pilot/controller workload. ATC clearance **must always** be received prior to flying a SID.

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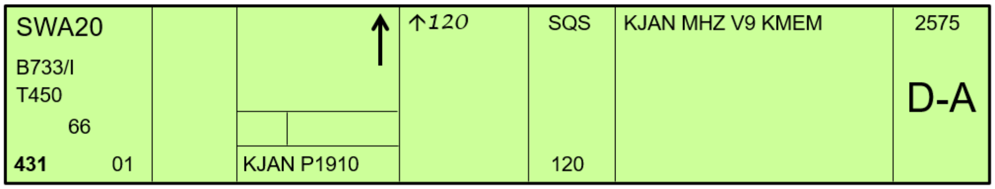
ROUTE ASSIGNMENT *(Continued)*

Route Use
JO 7110.65, par. 4-4-1

Phraseology Example

DESIGNATED AIRWAY



"Southwest Twenty, cleared to Memphis Airport via Victor Niner, climb and maintain one two thousand."

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⊙ Clear aircraft via routes consistent with the altitude stratum in which the operation is to be conducted by one or more of the following:

- Designated airways and routes

Phraseology

"VIA"

VICTOR (COLOR) (airway number) (the word "ROMEO" when RNAV)

or

J (route number) (word "ROMEO" when RNAV)."

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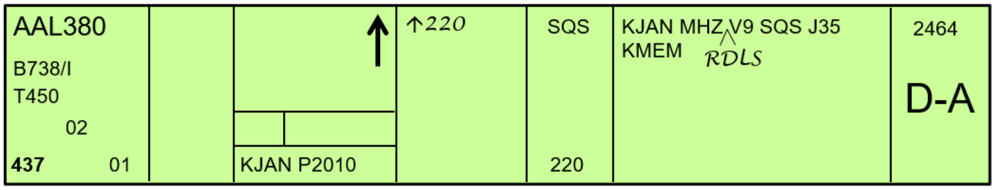
Route Use (Cont'd)
JO 7110.65, par. 4-4-1

DESIGNATED RADIALS OF AIRWAY



✈ **Phraseology Example**

"American Three Eighty, cleared to Memphis Airport, via radials of Victor Niner Sidon, J Thirty-Five, climb and maintain flight level two two zero."

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- Radials, courses, azimuths, or direct to or from NAVAIDs
  - To utilize an airway above/below its route structure
  - To define a route when an airway does **not** exist

✈ **Phraseology**

"DIRECT."

or

"VIA"

(name of NAVAID) (specified) RADIAL/COURSE/AZIMUTH

or

(fix) AND (fix)

or

RADIALS OF (airway or route) AND (airway or route)."

Continued on next page

Route Use (Cont'd)
JO 7110.65, par. 4-4-1

REROUTE ON DEPARTURE VIA RADIALS

✈ **Phraseology Example**

"Westwind Niner Zero Five Tango, cleared to Shreveport Airport via direct Sidon, the Sidon two four zero radial, until intercepting Monroe zero six zero radial Monroe Victor Eighteen, then as filed, climb and maintain one six thousand."

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- Radials, courses, azimuths, and headings of departure or arrival routes

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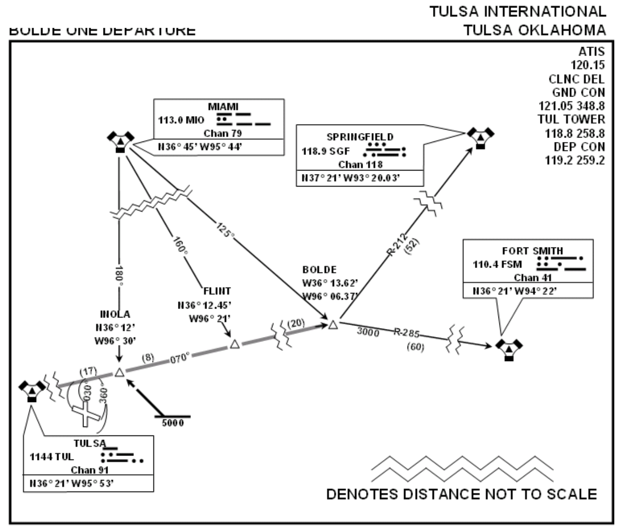
ROUTE ASSIGNMENT *(Continued)*

Route Use (Cont'd)
JO 7110.65, par. 4-4-1

BOLDE 1 DEPARTURE



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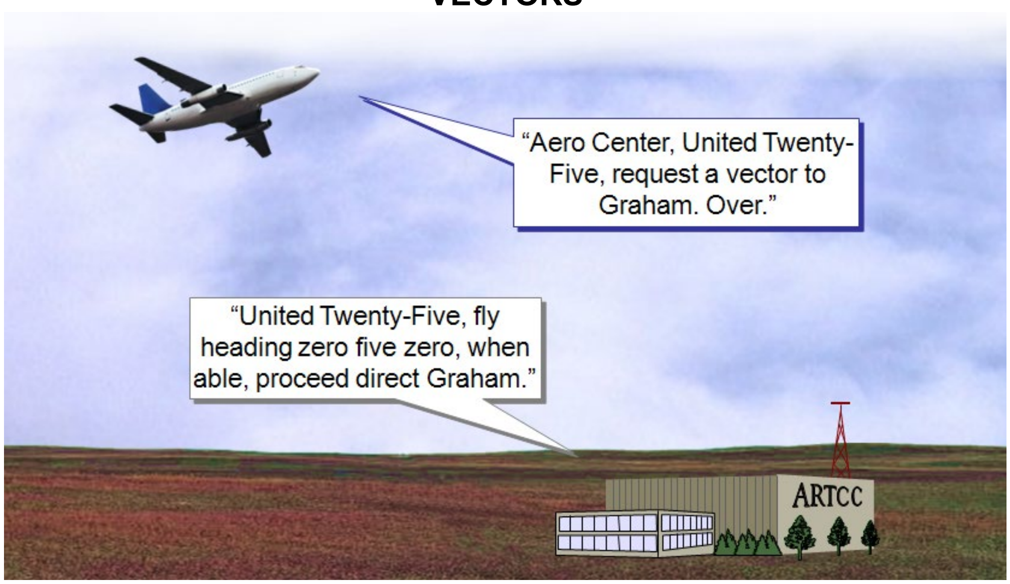
- Standard Instrument Departures (SIDs)/Standard Terminal Arrivals (STARs)/Flight Management System Procedures (FMSPs)

VECTORS

✈ **Phraseology Example**



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- Vectors

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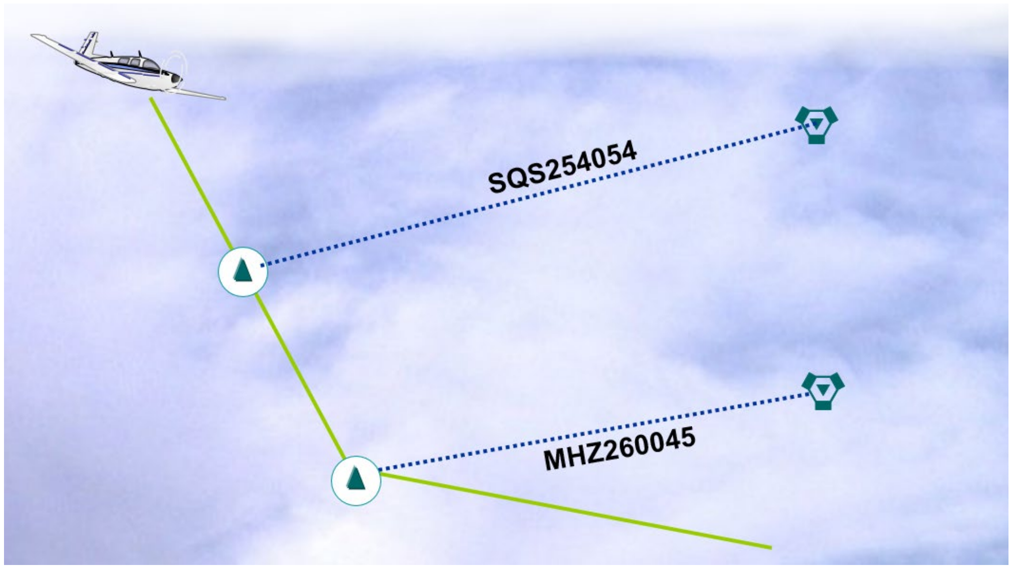
ROUTE ASSIGNMENT *(Continued)*

Route Use (Cont'd)
JO 7110.65, par. 4-4-1

- Degree-distance fixes for special military operations
- Courses, azimuths, bearings, quadrants, or radials within a radius of a NAVAID

FIX RADIAL DISTANCE FIXES



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- Fixes/waypoints defined in terms of:
  - Published name
  - Degree-distance from NAVAIDs
  - Latitude/longitude coordinates

✈
Phraseology

"DIRECT (fix/waypoint)." e.g., "DIRECT HEDUD"

"DIRECT TO THE (facility) (radial) (distance) FIX."

Example: "DIRECT TO THE MAGNOLIA TWO FOUR FIVE RADIAL THREE ZERO MILE FIX"

ALTITUDE STRATUM EXERCISE

Exercise

ALTITUDE STRATUM EXERCISE

Purpose: to practice determining which route to clear aircraft based on altitude stratum

Directions: answer the questions

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Directions

Record your answers to questions 1 – 7 in the space provided.

Questions

QUESTION 1

UAL35 A320/I T420 66	MCB 1500	28 15 MHZ	160	SQS		

QUESTION: Should this aircraft be cleared via victor airway or jet route? VICTOR

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Continued on next page

Questions (Cont'd)

QUESTION 2

N651CC C650/I T420 66	EIC 0256	22 03 17 MHZ	210✓	MEI		

◈ **QUESTION:** Should this aircraft be cleared via victor airway or jet route? JET

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QUESTION 3

N8010A GLF3/L T440 66	EIC	52 00 MHZ	230	MCB		

◈ **QUESTION:** Should this aircraft be cleared via victor airway or jet route? JET

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ALTITUDE STRATUM EXERCISE *(Continued)*

Questions (Cont'd)

QUESTION 4

N9051N AC80/A T250 66	GLH 1330	48 13 MHZ	160	MIZZE		

QUESTION: Should this aircraft be cleared via victor airway or jet route? VICTOR

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QUESTION 5

N56Q P28A/A T145 66	GLH 0022	48 00 MHZ	170	MIZZE	KLIT V74 MHZ V11 GCV KMSY/0139	

QUESTION: Is the route that the aircraft is assigned appropriate for the altitude stratum? YES

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Questions (Cont'd)

QUESTION 6

N64AP C500/A T280	MHZ 1911	21 19 DORTS	160	MLU	KSTF./.MHZ V417 MLU KRSN/1942 RDLS	

QUESTION: This aircraft requests climb to FL180, what would the new route clearance be?

CLEARED TO KRSN VIA AFTER MHZ THE RADIALS OF V417...

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QUESTION 7

DAL369 MD82/L T420	MEI 1534	44 15 MHZ	180		KATL./.MEI J20 MHZ J4 EIC KDFW RDLS	

QUESTION: Approaching Magnolia VORTAC, DAL369 requests clearance to one six thousand, what would the new route clearance be?

CLEARED TO DFW VIA AFTER THE RADIALS OF J4 BELCHER DIRECT

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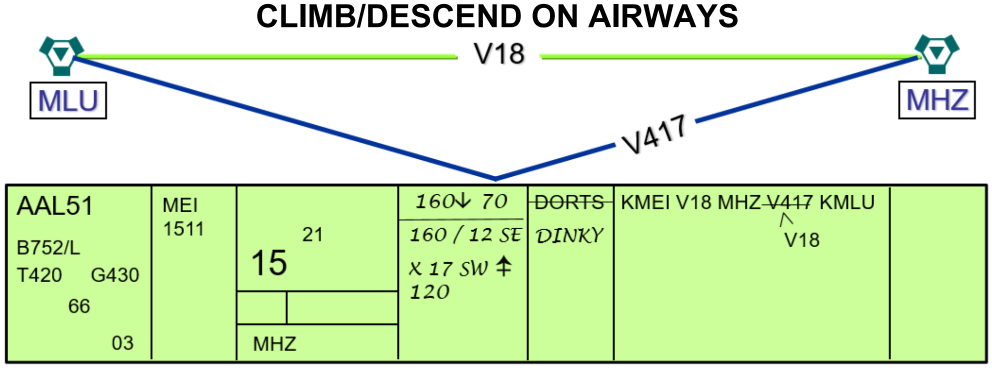
ROUTE STRUCTURE TRANSITIONS

Transition Within/ Between Route Structures
JO 7110.65, par. 4-4-2

⊙ To transition within or between route structures, clear an aircraft by one or more of the following methods, based on VOR, VORTAC, TACAN or MLS NAVAIDs:

- Vector aircraft to or from:
  - Radials
  - Courses
  - Azimuths of the airway or route assigned
- Assign a
  - Standard Instrument Departure (SID)
  - Standard Terminal Arrival (STAR)

CLIMB/DESCEND ON AIRWAYS



Phraseology Example

"American Fifty-One, cleared to DINKY Intersection, via after Magnolia, Victor Eighteen, maintain one six thousand until one two miles southeast Magnolia cross one seven miles southwest Magnolia VORTAC at or above one two thousand, descend and maintain seven thousand, hold northeast on victor eighteen expect further clearance one five two five."

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- Clear departing or arriving aircraft to climb or descend via:
  - Radials
  - Courses
  - Azimuths of the airway or route assigned

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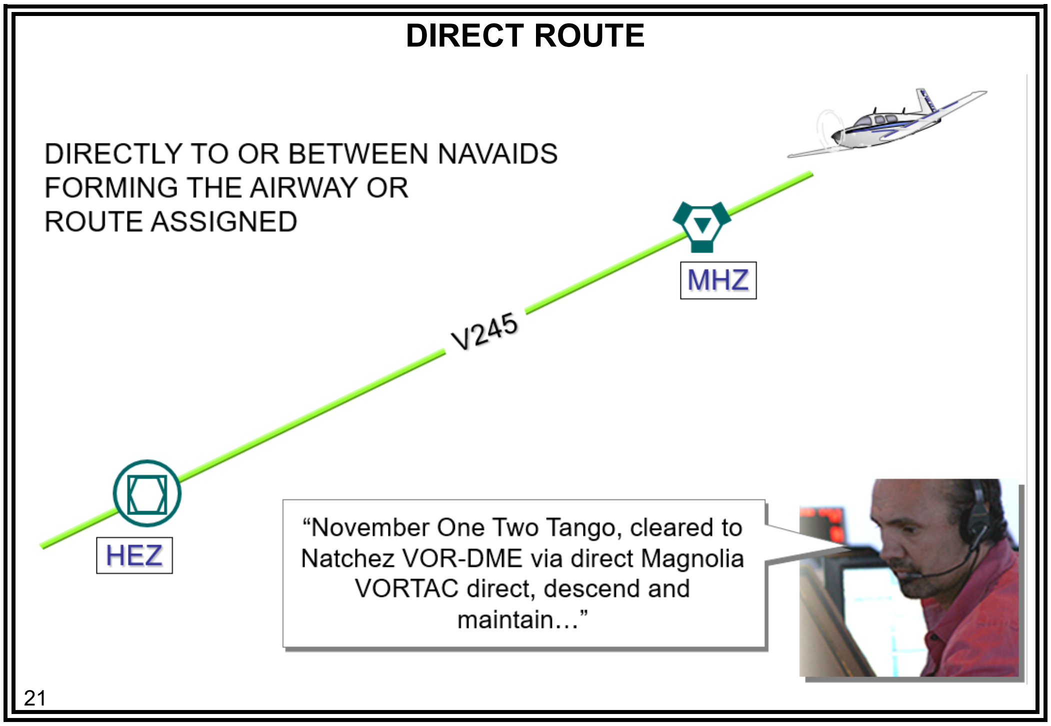
ROUTE STRUCTURE TRANSITIONS *(Continued)*

Transition Within/ Between Route Structures (Cont'd)
JO 7110.65, par. 4-4-2



- Clear departing or arriving aircraft directly to or between NAVAIDs forming the airway or route assigned.

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ROUTE STRUCTURE TRANSITIONS *(Continued)*

Transition Within/ Between Route Structures (Cont'd)
JO 7110.65, par. 4-4-2

✈
Phraseology Example

DESCENT ON AIRWAYS

AAL51 H/B772/L T475 G475 66 **675** 05	SQS 0902	11 09 MHZ ~~271005~~	240✓↓70 X35SE ↑ 120	MCB 0921	KMEM J35 KMCB SQS MHZ V9	6564

"American Fifty-One, after Sidon, cleared direct Magnolia Victor Niner McComb, cross three five miles southeast Magnolia VORTAC at or above one two thousand, descend and maintain seven thousand."

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- Clear aircraft to climb or descend via:
  - The airway or route on which flight will be conducted
  - Specified radials, courses, or azimuths of NAVAIDs

FIX RADIAL DISTANCE ACTIVITY

Activity

FIX RADIAL DISTANCE ACTIVITY

Purpose: to gain proficiency working with fix radial distances

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Description

In this activity, you will be presented with information related to the components that comprise a fix radial distance and then will be asked to answer a series of related questions. Feedback will be given immediately.

Directions

Access the IET eLearning menu. Select **Lesson 11 – Route Assignments**. Click on the title to launch the **Fix Radial Distance** activity.

Time Allotted

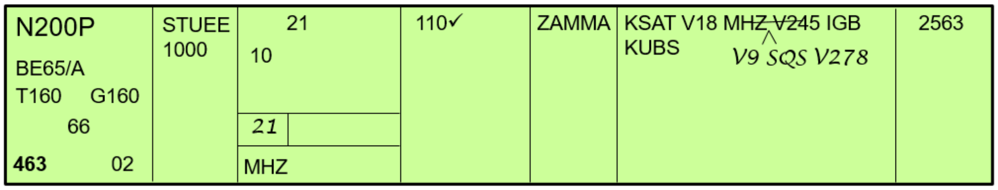
30 minutes

CLEARANCE AMENDMENTS

Route/Altitude Amendments
JO 7110.65, par. 4-2-5

✈
Phraseology Example

ROUTE AMENDMENT EXAMPLE 1



"Beechcraft Two Zero Zero Papa, change Magnolia Victor Two Forty-Five Bigbee to read Magnolia Victor Niner Sidon Victor Two Seventy-Eight Bigbee direct Columbus."

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- Amend route of flight in a previously issued clearance by one of the following methods:
  - State which portion of the route is being amended and then state the amendment

✈
Phraseology

"CHANGE (portion of route) TO READ (new portion of route)."

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CLEARANCE AMENDMENTS *(Continued)*

Route/Altitude Amendments (Cont'd)
JO 7110.65, par. 4-2-5

✈ **Phraseology Example**

ROUTE AMENDMENT EXAMPLE 2

N200P BE65/A T160 G160 66 **462** 05	HLI	11 07 *11* SQS	110✓	MHZ	KSTL ./. HLI V535 SQS ~~V9~~ *V557* MHZ V11 KGCV	2571

"Beechcraft Two Zero Zero Papa, cleared via after Sidon Victor Five Fifty-Seven Magnolia, rest of route unchanged."

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NOTE: If "rest of route unchanged" is left out, the pilot may be confused as to what routing to fly after MHZ or the pilot may mistakenly assume that the clearance limit is MHZ.

- State the amendment to the route and then state that the rest of the route is unchanged

✈ **Phraseology**

"(Amendment to route), REST OF ROUTE UNCHANGED."

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CLEARANCE AMENDMENTS *(Continued)*

Route/Altitude Amendments (Cont'd)
JO 7110.65,
par. 4-2-5

ROUTE AMENDMENT EXAMPLE 3

SWA632 B733/A T450 G435 66 **211** 01	HEZ 1100	11 11 *11* MHZ	110✓	SQS	KMSY HEZ ~~V245~~ MHZ V555 SQS KGWO	1362

✈
Phraseology Example

"Southwest Six Thirty-Two, cleared direct Magnolia VORTAC."

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- Issue a clearance "direct" to a point on the previously issued route

✈
Phraseology

"CLEARED DIRECT (fix)."

NOTE: Clearances authorizing "direct" to a point on a previously issued route do **not** require the phrase, "rest of route unchanged." However, it **must** be understood where the previously cleared route is resumed. When necessary, "rest of route unchanged" may be used to clarify routing.

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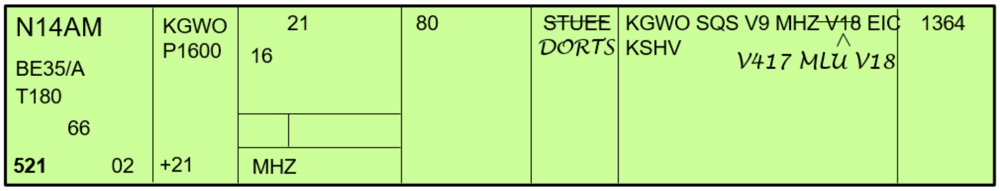
CLEARANCE AMENDMENTS *(Continued)*

Route/Altitude Amendments (Cont'd)
JO 7110.65, par. 4-2-5

ROUTE AMENDMENT EXAMPLE 4



✈ **Phraseology Example**

Bonanza One Four Alpha Mike has been cleared to the Shreveport (SHV) Airport via Sidon Victor Niner Magnolia Victor Eighteen Belcher (EIC) Direct Shreveport Airport Climb and Maintain Eight Thousand. After takeoff, the aircraft is rerouted via after Magnolia Victor Four Seventeen Monroe Victor Eighteen Belcher.

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- Issue the entire route by stating the amendment

✈ **Phraseology**

The Controller can issue **any** one of the following clearances:

- "Bonanza One Four Alpha Mike **Change** Victor Eighteen Belcher **To Read** Victor Four Seventeen Monroe Victor Eighteen Belcher."
- "Bonanza One Four Alpha Mike cleared via after Magnolia Victor Four Seventeen Monroe Victor Eighteen Belcher, **Rest of Route Unchanged**."
- "Bonanza One Four Alpha Mike Cleared via after Magnolia Victor Four Seventeen Monroe Victor Eighteen Belcher **Direct Shreveport Airport.**"

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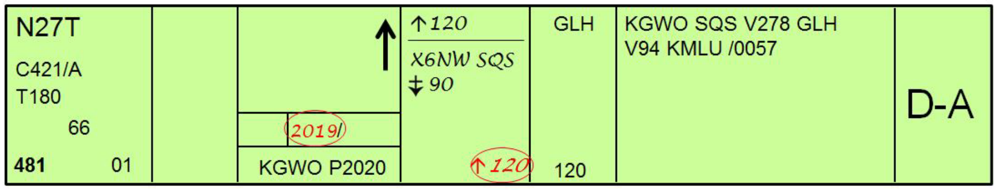
CLEARANCE AMENDMENTS *(Continued)*

Route/Altitude Amendments (Cont'd)
JO 7110.65, par. 4-2-5

ROUTE/ALTITUDE AMENDMENT
ORIGINAL CLEARANCE



✈ **Phraseology Example**

"Cessna Two Seven Tango, cleared to Monroe Airport via direct Sidon Victor Two Seventy Eight Greenville Victor Ninety Four. Cross six miles northwest Sidon VORTAC at or below niner thousand. Climb and maintain one two thousand."

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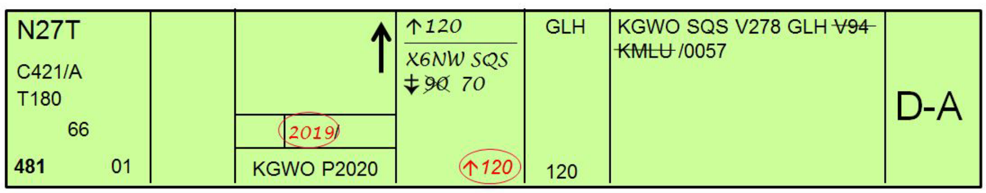
ROUTE/ALTITUDE AMENDMENT (CONT'D)
AMENDED ROUTE AND ALTITUDE RESTRICTION



✈ **Phraseology Example**

"Cessna Two Seven Tango, cleared to Greenville Airport via direct Sidon Victor Two Seventy-Eight. Cross six miles northwest Sidon VORTAC at or below seven thousand. Climb and maintain one two thousand."

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⊙ When route or altitude in a previously issued clearance is amended, restate all applicable altitude restrictions.

Continued on next page

Route/Altitude Amendments (Cont'd)
JO 7110.65, par. 4-2-5

NOTE: Restating previously issued altitude to "maintain" is an amended clearance. If altitude to "maintain" is changed or restated, whether prior to departure or while airborne, and if previously issued altitude restrictions are omitted, altitude restrictions are canceled, including SID/STAR/ATC altitude restrictions, if any.

Example: A Greenwood Departure is cleared to cross one seven miles southwest Sidon VORTAC at or below five thousand, cross one seven miles northwest Magnolia VORTAC at or above seven thousand, climb and maintain one zero thousand. Shortly after takeoff the aircraft request a final altitude of eight thousand. Because the altitude restriction remains in effect, the controller issues an amended altitude clearance as follows:

"Amend Altitude. Cross one seven miles southwest Sidon VORTAC at or below five thousand, cross one seven miles northwest Magnolia VORTAC at or above seven thousand, climb and maintain eight thousand."

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CLEARANCE AMENDMENTS *(Continued)*

Knowledge Check

KNOWLEDGE CHECK

QUESTION: What are the four ways to amend a route of flight?

CHANGE PORTION OF ROUTE TO READ NEW PORTION OF ROUTE BAD (AMENDMENT) REST OF ROUTE UNCHANGED

CLEARED DIRECT (FIX)

ALTITUDE TO MAINTAIN

30

KNOWLEDGE CHECK

N23GP LJ24/I T450 66		↑ 1733 /1733 KGWO P1730	↑120 X17SE SQS ↓70 X17NW MHZ ↑90	MHZ 130	KGWO SQS V9 MCB KBTR MHZ V557	D-A

QUESTION: N23GP was cleared as filed. Later, the aircraft was rerouted via SQS V9 MHZ V557 MCB KBTR with the same restrictions and altitude. How should the clearance be amended?

A. Amend route and say, “Altitude and restrictions remain unchanged.”

B. Amend route only.

C. Amend route and restate applicable altitude restrictions.

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IN CONCLUSION

Lesson Review

LESSON REVIEW

The following topics were covered in this lesson:

– Clearance and route assignment procedures for IFR aircraft
– Phraseology for issuing selected clearances

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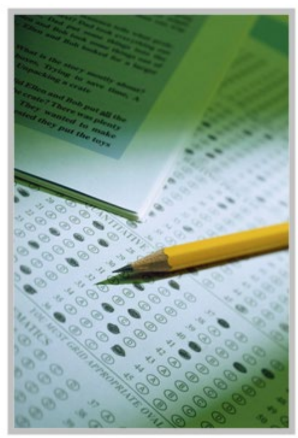
End-of-Lesson Test

END-OF-LESSON TEST

Route Assignments



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